



Carbonate pore system evaluation using the velocity–porosity–pressure relationship, digital image analysis, and differential effective medium theory



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ABSTRACT

Carbonate reservoirs exhibit heterogeneous pore systems and a wide variety of grain types, which affect the rock's elastic properties and the reservoir parameter relationships. To study the Albian carbonates in the Campos Basin, a methodology is proposed to predict the amount of microporosity and the representative aspect ratio of these inclusions. The method assumes three pore-space scales in two representative inclusion scenarios: 1) a macro–mesopore median aspect ratio from the thin-section digital image analysis (DIA) and 2) a microporosity aspect ratio predicted based on the measured P-wave velocities. Through a laboratory analysis of 10 grainstone core samples of the Albian age, the P- and S-wave velocities (V_p and V_s) are evaluated at effective pressures of 0–10 MPa. The analytical theories in the proposed methodology are functions of the aspect ratios from the differential effective medium (DEM) theory, the macro–mesopore system recognized from the DIA, the amount of microporosity determined by the difference between the porosities estimated from laboratorial helium-gas and the thin-section petrographic images, and the P-wave velocities under dry effective pressure conditions. The DIA procedure is applied to estimate the local and global parameters, and the textural implications concerning ultrasonic velocities and image resolution. The macro–mesopore inclusions contribute to stiffer rocks and higher velocities, whereas the microporosity inclusions contribute to softer rocks and lower velocities. We observe a high potential for this methodology, which uses the microporosity aspect ratio inverted from V_p to predict V_s with a good agreement. The results acceptably characterize the Albian grainstones. The representative macro–mesopore aspect ratio is 0.5, and the inverted microporosity aspect ratio ranges from 0.01 to 0.07. The effective pressure induced an effect of slight porosity reduction during the triaxial tests, mainly in the microporosity inclusions, slightly changing the amount and the aspect ratio of the microporosity.

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1. Introduction

Carbonate rocks have a great economic significance and hold more than 50–60% of the oil and gas reserves worldwide (e.g., Burchette, 2012). In Brazil, such reservoirs represent a significant portion of the deepwater oil production, whose importance has increased with the recent discoveries in the post- and pre-salt oil deposits (Bruhn et al., 2003). Those rocks commonly display heterogeneities due to diagenesis and exhibit complicated mineral composition, pore structure, and texture variations that may cause a low hydrocarbon recovery (Anselmetti and Eberli, 1993, 1999; Xu et al., 2007). For example, for a given reservoir with a porosity of ~25%, the permeability can vary by more than four orders of magnitude due to the pore structure changes. The pore type variations can also produce the seismic velocity changes

at a given porosity (Sun et al., 2006), a behavior similar to that observed in the Albian carbonates in Brazil.

The relationship between reservoir rocks and elastic properties is important to understanding and improving the practical rock physics models. In this context, the bulk and shear moduli of the dry-rock framework are two essential parameters for fluid substitution and S-wave prediction, for example. The elastic moduli of a dry rock depends not only on the porosity but also on the pore geometry (Li and Zhang, 2011). Carbonate rocks have pore systems composed of intergranular (interparticle) and intercrystalline primary porosity (Anselmetti and Eberli, 1999). The secondary porosity commonly involves inclusions of oomoldic, moldic, and vuggy pores, which are considered to be rounded and enhance the rock stiffness compared with the interparticle pore, inducing a faster seismic wave propagation, whereas microporosity and fractures tend to be flat and make the rock softer (Berryman, 1995; Kumar and Han, 2005; Wang et al., 2009; Xu and Payne, 2009; Zhan et al., 2012).

Many rock physics studies assume the pore aspect ratio of inclusions, the main textural parameter that contributes to the stiffness or softness

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of a rock, is a geometric variable that influences the acoustic velocities (Agersborg et al., 2005; Assefa et al., 2003; Kumar and Han, 2005; Rossebø et al., 2005; Saleh and Castagna, 2004). For carbonate reservoirs that are dominated by the secondary porosity, Eberli et al. (2003) discussed the velocity–porosity relationship for rocks with different pore systems, including microporosity, moldic, interparticle, and densely cemented rocks. According to the theoretical concept, high-aspect-ratio pores, such as molds and vugs, provide more grain-to-grain contact than interparticle and intercrystalline pores, thus decreasing the pore compressibility and providing more stiffness to the rock with an equal porosity (Mavko and Mukerji, 1995; Saleh and Castagna, 2004; Weger et al., 2009).

The Digital Image Analysis (DIA) methodology can be applied to evaluate mineral structures and pore systems (Anselmetti and Eberli, 1999; Anselmetti et al., 1998). It can be used to differentiate the pore space from a matrix material and quantify the pore structure properties, such as size, shape, distribution of grains, cementation, and porosity. The pore shape and pore network complexity have a strong influence on the permeability and the values of acoustic velocities (Berryman and Blair, 1987; Mavko et al., 1998). Weger (2006) and Weger et al. (2009) showed that carbonate rocks have pore structures constructed of macro-, meso-, and micropores. Macro- and mesopores can be detected in thin-section images, and the amount of microporosity is calculated as the difference between the observed porosity in DIA and the measured porosity from core samples. Furthermore, the microporosity can be studied by applying an X-ray microtomography or other methods of more accurate resolution, such as a nanotomography, which contributes to a better quantification.

In this work, 10 grainstone core samples from the Albian age were studied, and laboratorial measurements were performed. The goal is to predict the microporosity parameters using the velocity–porosity–pressure relationship, thin-section image analysis, and the rock physics differential effective medium (DEM) theory under ultrasonic and dry conditions. The method considers three pore-space scales in two representative inclusion scenarios: 1) the macro–mesopore median aspect ratio from DIA and 2) the microporosity aspect ratio predicted by the measured P-wave velocity.

2. Geological setting

The Albian carbonate reservoirs in the Campos Basin are located in southeastern Brazil and form part of the NE-trending, elongated shoals (up to 20 m thick, less than 1 km wide, and up to 2.5 km long), which are composed mostly of grainstones and packstones containing oncolites, peloids, oolites, and rare bioclasts (Fig. 1). The carbonate shelf cycles consist of upward shoaling lithological sequences starting with peloidal wackestones, followed by oncolitic/oolitic packstones and oncolitic/oolitic grainstones. This series represents the common facies associations. Most of the deposition occurred during the marine regression periods. According to Bruhn et al. (2003), between the shoals, finer-grained carbonates, particularly peloidal calcisiltites, were deposited in a lower energy environment. The calcarenites and calcirudites form the oil reservoirs in shallow water (100–200 m) and are composed mostly of grainstones and packstones containing oncolites, peloids, oolites, and rare bioclasts. Thus, two megasequences define the Albian carbonates: a shallow carbonate platform (Early to Middle Albian) and a marine transgressive megasequence (Late Albian to Early Tertiary). In this study, the Early to Middle Albian grainstone core-plug samples were selected from two cored wells. The matrix-free and oncolite/oolite-rich calcarenites comprise the best reservoir facies, with the porosities ranging between 20 and 34% and the permeabilities exceeding 100 mD.

The set of selected samples includes the oncolite/oolite grainstones with a good porosity (~24.8%) and permeability (~4–222 mD) from Well 1 and the oncolite/oolite grainstones with calcite cementation causing a reduction of the permeability (~0.88–2.03 mD) from Well 2,

totaling 10 core-plugs (1.5" [38.1 mm] in diameter by 2" [50.8 mm] in length). Thin-sections were taken from the top of each plug sample before performing any laboratorial measurement. The porosities were measured by helium-gas injection and the permeabilities by nitrogen-gas injection, with the experimental uncertainties approximately 0.3% and 5–10 mD, respectively. The mineralogical characterizations were performed using an X-ray diffractogram (XRD) analysis and the Rietveld method on representative pieces of the core-plug samples. The accuracy of the combination of both the analysis and the method was approximately $\pm 3\%$, and the minimum detectable amount was approximately 0.3% of the sample weight. The results showed that calcite predominates and that noncarbonate minerals account for less than 5% of the samples (see Archilha et al., 2013 and Lima Neto et al., 2013). The matrix mineral moduli were predicted with the measured mineral weight percentage and Voigt–Reuss–Hill average method (Mavko et al., 1998), except the W1-Im07 and W1-Im08 samples, which were calculated using a linear regression based on the similarity of the textural and density values from the samples at depth and calibrated with the P- and S-wave velocity measurements. The experimental error on velocities was approximately 1%. The data set is listed in the Appendix A.

3. Differential effective medium (DEM) theory

This model assumes isolated pores embedded in a host material that remains continuous at all porosities. The DEM theory simulates the porosities in a composite medium of two phases by incrementally adding small amounts of pores (phase 2) into a matrix (phase 1) until the total porosity (ϕ) is attained (Berryman, 1992):

$$(1-\phi) \frac{d}{d\phi} [K^*(\phi)] = (K_2 - K^*) P^{(*2)}(\phi), \quad (1)$$

$$(1-\phi) \frac{d}{d\phi} [\mu^*(\phi)] = (\mu_2 - \mu^*) Q^{(*2)}(\phi), \quad (2)$$

where $K^*(0) = K_1$ and $\mu^*(0) = \mu_1$ are the bulk and shear moduli of the host material, respectively (phase 1); K_2 and μ_2 are the bulk and shear moduli of the inclusions, respectively, with $K_2 \approx 0$ and $\mu_2 = 0$ for a dry rock. The terms $P^{(*2)}$ and $Q^{(*2)}$ are geometric factors that depend on the aspect ratio of the inclusions, as predicted in Appendix B. The P- and S-wave velocities can be calculated using the results of bulk and shear moduli.

4. Methodology

In this study, we use the DEM theory to predict the microporosity inclusion aspect ratios to characterize the complex constituents and pore geometries with the elastic properties of the Albian carbonates. After establishing the geological setting with the measurements and mineral information, the physical properties of the Albian grainstones can be linked to the DIA parameters through thin-section micrographs, allowing for the characterization of macro- and mesopores. The rock is characterized by macro-, meso-, and micropore systems, which are expressed in the physical properties as P- and S-wave velocities and correlated with bulk (K) and shear (μ) moduli. The goal of this study is to determine the best microporosity aspect ratio that can be combined with the macro–mesopore aspect ratios, which are fractions of the pore inclusions and are predicted from the DIA and pore system modeling through a DEM theory inversion (Fig. 2). The method considers three pore-space scales in two representative inclusion scenarios: 1) the macro–mesopore median aspect ratio and 2) the microporosity aspect ratio predicted by the measured P-wave velocity.

The theory of poroelasticity predicts an ability of rocks to resist and recover from the deformations produced by the induced external forces. Therefore, when an effective pressure is applied to a drained core

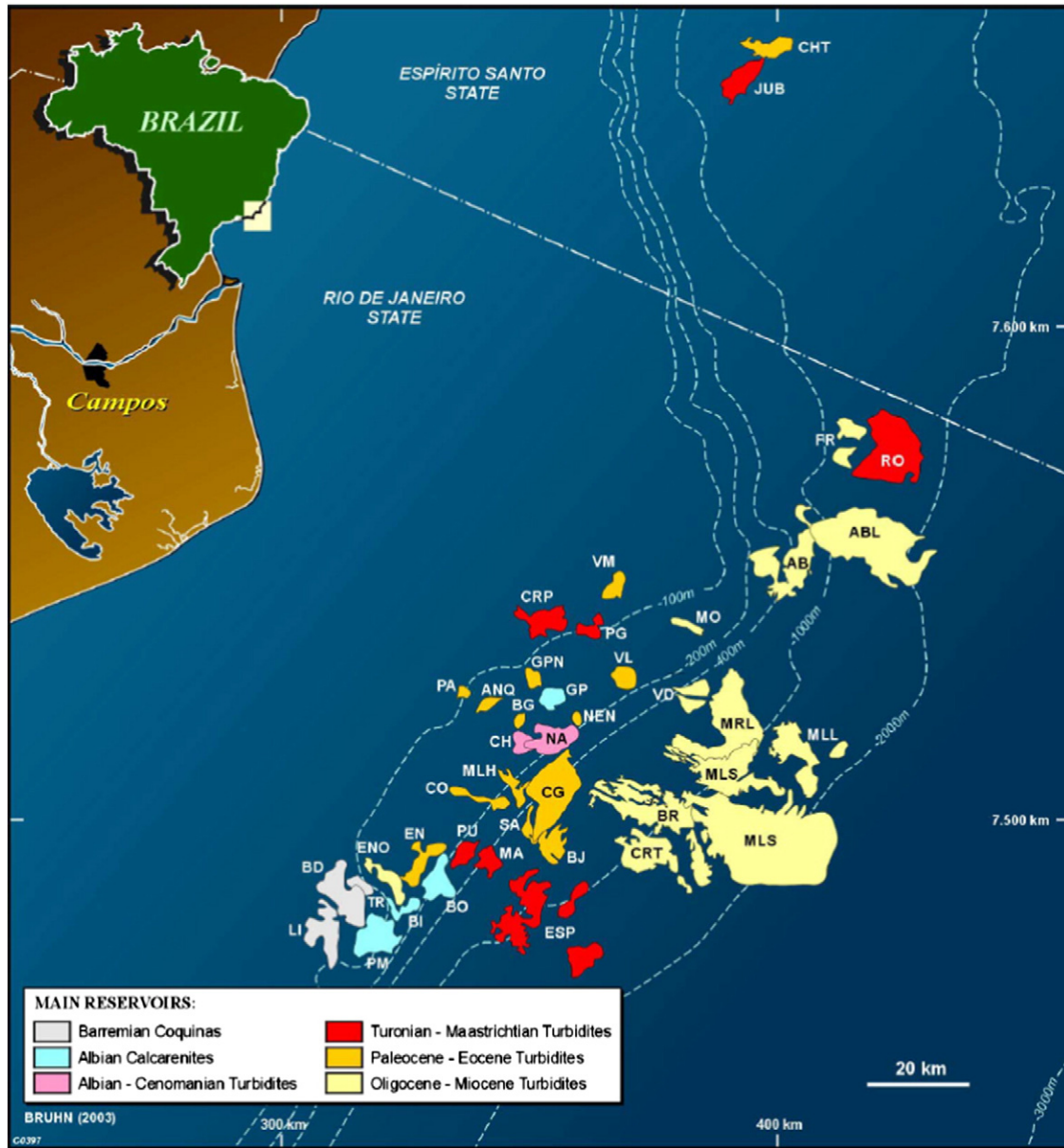


Fig. 1. Location map of the oil fields in the Campos Basin. The Albian calcarenite reserves are shown in light blue (Bruhn et al., 2003).

sample during a hydrostatic loading test, there is a null pore pressure condition, and the core sample exhibits a higher volume deformation in contrast to the undrained case. In fact, if a rock is subjected to a pressure regime, the deformation of crystals is the mechanical property of the smaller variations, though the rock pore space is sensitive to the pressure effects and structural discontinuities, such as micropores, fractures, and voids, that cause a volumetric deformation (Fjaer et al., 2008; Jaeger et al., 2007). Here, it was assumed that the volumetric reduction is caused by the microporosity diminishing when an effective pressure is applied, which leads to an increase in the P-wave velocity by increasing the predominance of rounded macro-mesopores and reducing the bulk porosity. The method depicted in Fig. 2 was applied under effective pressure conditions and the porosity was recalibrated using the measured volumetric reduction.

5. Assumptions, limitations, and considerations

In this study, the DEM theory in the methodology assumes high frequencies, such as ultrasonic P- and S-wave velocities, measured

from the drained core samples during triaxial laboratory experiments. However, for the low-frequency as on a seismic scale, an approach for upscaling is to apply the DEM theory by inserting dry inclusions into the predicted drained rock frame and to perform a fluid saturation as the final step using the Gassmann's equation that assumes seismic conditions (see the Xu-White and Xu-Payne models (Xu and White, 1995; Xu and Payne, 2009)). In addition, the saturated carbonates with interparticle and intergranular primary porosities can be P-wave velocity modeled using the Wyllie's time-average equation (Anselmetti and Eberli, 1999), although for dry conditions, we can use the DEM theory with representative interparticle aspect ratio inclusions (Xu and Payne, 2009), which is a reference line for the oncolite/oolite Albian grainstones. Another upscaling approach can be considered by applying the Backus' average (Backus, 1962) when the wavelength is large compared with the layer thickness at lower frequencies (seismic scale, e.g.) from high or very-high frequencies.

Despite the assumptions and limitations of the rock physics models described in the literature, it is common to use the calibration of parameters to study methodologies aiming to predict the properties of interest

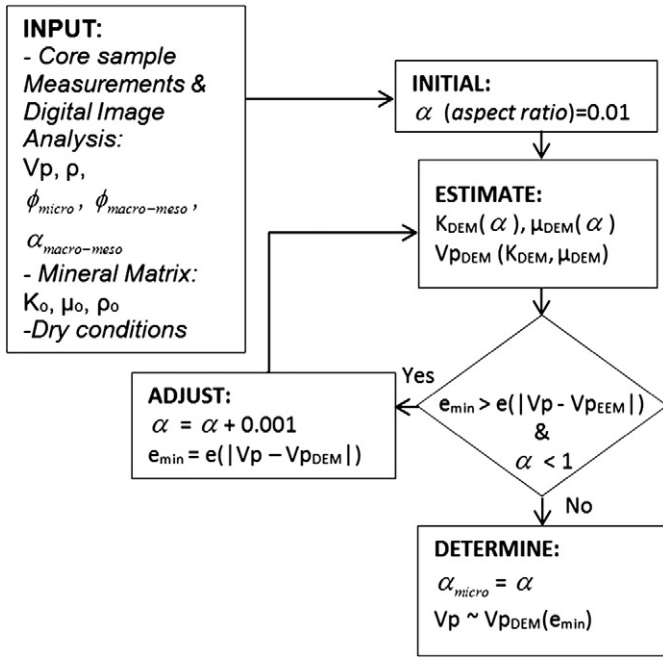


Fig. 2. The methodology proposed to predict the micropore aspect ratio (α_{micro}).

and comparing the results with prior knowledge. The methodology proposed in this study uses the DEM model, an effective elastic media theory including the bounds and mixing laws (Mavko et al., 1998), considering the idealized ellipsoidal pore inclusion shapes that are statistically estimated from the DIA study, preserving the pore characteristics identified as area and elongation. The DIA study was incorporated to reduce the limitations with the idealized pore shapes and provide a better rock characterization. Thus, different carbonate rocks could be evaluated by applying the methodology in this work by changing the DEM model for another adequate effective elastic media theory as the Kuster–Toksöz formulation and the Self-consistent approximations of effective moduli. Our previous study (see Lima Neto et al., 2013) showed that the DEM model is suitable for predicting the elastic moduli and velocities of the oncolite/oolite Albian grainstones.

6. Data analysis and results

6.1. Digital image analysis (DIA) procedure

Textural parameters can be calculated from a DIA procedure according to Weger (2006). DOMsize provides an indication of a sample's dominant pore-size range, which is defined as the upper boundary of the pore sizes of which 50% of the porosity in a thin-section is composed. PoA is the ratio between the total pore-space area of a thin-section and the total perimeter that encloses the pore space. Generally, a small PoA value indicates a simple pore system. Gamma (γ) describes the roundness of pores as the perimeter over an area of an individual pore normalized to a circle, as proposed by Anselmetti et al. (1998). A perfect round circle would have $\gamma = 1$. Aspect ratio (α) is the ratio between the major and minor semiaxes of an ellipse that encloses the pore, and it describes the elongation of the pore-bounding ellipsoid. The α can be estimated for macro-mesopore systems using the median of pores recognized from the thin-sections. For a spherical pore, $\alpha = 1$. In this study, the ellipsoids recognized from DIA are considered oblate ($\alpha < 1$). The parameters are classified into two different types: global parameters that describe the pore system recognized on the thin-section images (DOMsize, PoA, and

f_{micro}) and local parameters from individual pores (γ and α) (Castro and Rocha, 2013; Russ, 1998; Weger, 2006; Weger et al., 2009).

The thin-sections from the core samples were impregnated with blue epoxy, and micrographs were taken using an optical petrographic microscopy with a resolution of 764×574 pixels and magnifications between $25 \times$ ($\sim 16.32 \mu\text{m}^2/\text{pixel}$) and $200 \times$ ($\sim 0.28 \mu\text{m}^2/\text{pixel}$), to show the textural complexities, pore fabrics, and interparticle pore predominance (Fig. 3). The images were acquired using non-polarized light, and small noises were eliminated from the analysis by spatial filter applications, such as a low-pass filter that attenuates the color gradients, and a median filter that eliminates the small discontinuities without any geometrical consequence (Damiani et al., 2000).

After the filtering process, the thin-section digital images were stored as 24 bit files in the RGB color model for segmentation, which includes a neural network method for pattern recognition and assisting the geometrical characterization of the pore microstructures. Thresholding was performed using an inter-class variance maximization method (Coster and Charmant, 1989), applied to the histogram of each color component by the automated and manual information on saturation and intensity given during the analysis. The spectrum of pores was displayed in 2D binary images, and the pore size distribution was obtained by successive opening, which was derived from mathematical morphology and using balls with increasing radius (Coster and Charmant, 1989). The resulting image can be viewed as the union of balls completely enclosed in the porous phase. As a result, an octagonal ellipse represents each pore recognized, keeping approximately the same pixel area and elongation (Fig. 3(E)), and the pore aspect ratio (α) inclusions are considered oblate ($\alpha < 1$) due to the lack of an orientation reference from the 2D images (Fig. 3(F)). Details about the filtering and segmentation processes adopted in this work are in Damiani et al. (2000).

Textural parameters were calculated from the thin-section image analysis to quantify the pore structures and to describe several aspects of the pore system. We assumed that the resolution of the optical petrographic microscopy could not characterize microporosity but only macro- and mesopores. These types of porosity are associated with interparticle and spherical pores and are often dominant in grainstones. Therefore, we also assumed that the pore system described by the thin-section can be representative of the core plug. Consequently, the amount of microporosity (f_{micro}) and microporosity (ϕ_{micro}) can be calculated as the difference between the observed macro-mesoporosity in the DIA ($\phi_{\text{thin-section}}$) and the measured helium-gas porosity from the core samples ($\phi_{\text{helium-gas}}$), assuming that the helium-gas can fill the whole connected pore system (Fig. 4). The following equations describe this relationship:

$$\phi_{\text{micro}} = \phi_{\text{helium-gas}} - \phi_{\text{thin-section}}, \quad (3)$$

$$\phi_{\text{thin-section}} = \phi_{\text{macro-meso}}, \quad (4)$$

$$f_{\text{micro}} = \frac{\phi_{\text{micro}}}{\phi_{\text{helium-gas}}}, \quad (5)$$

$$f_{\text{macro-meso}} = 1 - f_{\text{micro}}. \quad (6)$$

Appendix A.3 shows the magnification acquisition values for each thin-section evaluated, including a set of all magnifications, as well as a statistical representation of the pore system. Fig. 5 shows the macro-mesopore aspect ratio balance and the representative median value for each core sample from Wells 1 and 2. We find $\alpha = 0.5$ to be a representative aspect ratio value for the Albian grainstone core samples studied.

The PoA ratio tends to increase with magnification, and this global parameter is highly dependent on the image resolution due to the

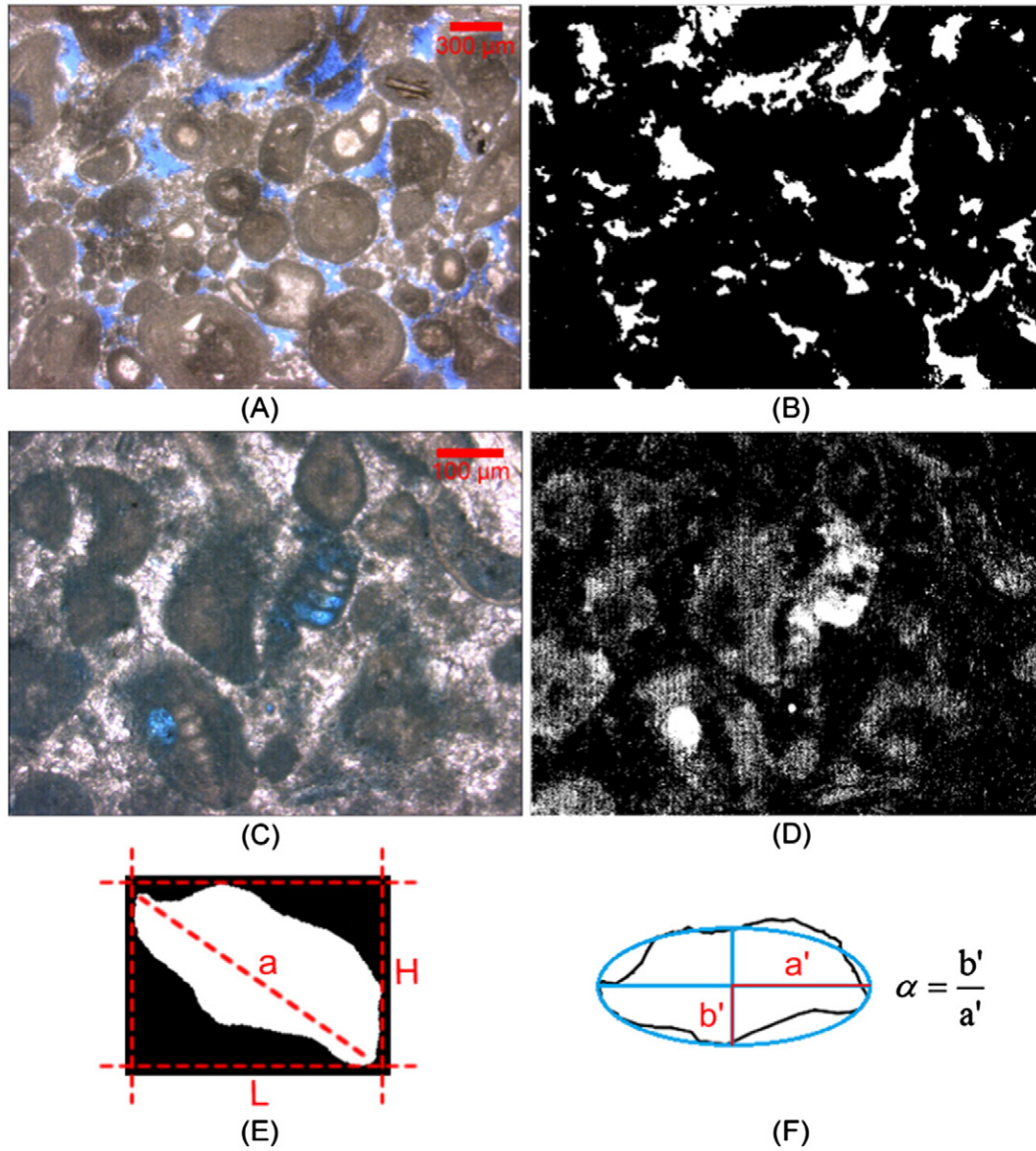


Fig. 3. The thin-sections (A and C) taken from the core samples (W1-1m1 and W2-1m1, respectively) showing the complex texture and pore fabric and the respective binary images (B and D) from the DIA methodology. The main textures from the core samples are oncolite/oolite grainstones: Well 1 (A, B) and Well 2 (C, D) – calcite cementation occurrence. The pores recognized (E) are measured in pixel and approximated to the octagonal ellipsis as oblate pore aspect ratio (α) inclusions (F).

detection of perimeter and shape in more details. In this case, we considered that the macro–mesopore system parameters DOMsize, f_{micro} , γ , and α can be characterized using all of the magnifications available, while PoA should be evaluated in the core sample images with the minimal magnification (25 $\times$). The global and local parameters were combined with the velocity measurements (Section 6.2) aiming to predict microporosity aspect ratio (Section 6.3), considering that the pore system is composed of macro–mesopores and microporosity.

6.2. Velocity measurements

The ultrasonic P- and S-wave velocities are measured in the room-dried samples at effective pressures (0–10 MPa) in triaxial tests, using one P-wave (1.3 MHz) and two independent orthogonally polarized S-wave (900 kHz) piezoelectric transducers simultaneously via the pulse transmission technique. The axial stress is calculated using load cell measurements and the sample initial diameter (measured with a caliper). The axial deflection is measured by LVDTs (Linear Variable Differential Transformers). The confining pressure is measured by

pressure transducers, and the radial deflection measurements are performed using a cantilever transducer. The volumetric reduction of the core samples is evaluated by the axial and radial deflection values at the effective pressure induced during the triaxial tests. The Albian carbonate core samples have heterogeneities, such as vugs and microfractures, that cause brittleness; therefore, we limited the effective pressures to 10 MPa for the dry condition triaxial tests (Fig. 6).

According to the proposed methodology, we assume that the volumetric reduction is caused by the closing of micropores when an effective pressure is applied. This process leads to an increased P-wave velocity through the increased predominance of rounded macro–mesopores and a reduction in porosity. Fig. 6 shows the scattering of P-wave velocities versus porosity under the effective pressure conditions. We see a slight porosity reduction (~ 0.4 – 3.6%) with the increase in pressure (Fig. 6(A)), and the impact on microporosity (Fig. 6 (B and C)). Spherical, interparticle, and microcrack pore geometries with aspect ratios of 1.0, 0.1, and 0.01, respectively, were calculated using the DEM theory and showed in Figs. 6, 7 and 9 as the reference dashed lines for the dry and clean calcite limestone in the

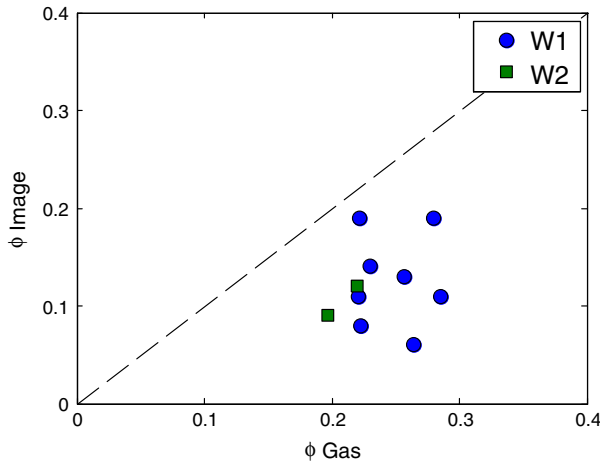


Fig. 4. The porosity difference evaluated using Digital Image Analysis (DIA) versus helium-gas measurements. Helium-gas porosity was assumed to access the whole connected pore system, including microporosity.

Vp- ϕ crossplots. The interparticle line is the main reference for the oncolite/oolite Albian grainstones as expected, and the Vp- ϕ scattering reflects an increase in the rounded inclusions (tending towards the spherical line direction) or the microporosity inclusions (tending towards the microcrack line direction) by the fraction and aspect ratio balances of the geometric inclusions.

6.3. Prediction of the microporosity aspect ratio and S-wave velocity

As a result of the DIA methodology, the microporosity inclusion aspect ratio was predicted and analyzed for three proposed case conditions using the following:

- 1) the macro-mesopores from all magnification data set (Appendix A.3 and A.4) and a Vp measured at each effective pressure (Fig. 6);

- 2) the macro-mesopores from all magnification data set (Appendix A.3 and A.4) and a Vp measured at an effective pressure of 3.5 MPa (Figs. 7 and 8); and,
- 3) the macro-mesopores from the 25 $\times$ magnification data set (Appendix A.3) and a Vp measured at an effective pressure of 3.5 MPa (Fig. 9).

The results from the case condition #1 express the effect of effective pressure on the pore system evaluated for each core sample (Fig. 6). The P-wave velocity increases with a high effective pressure, and we verify a volume reduction in the core samples during the experiments. This pattern leads to the conclusion, according to the proposed methodology, that a slight porosity volume reduction is caused by the effective pressure loading (Fig. 6(A)), which causes a reduction in the microporosity fraction and a slight increase in the microporosity aspect ratio as predicted (Fig. 6 (B and C)). This conclusion is explained by the closing of flat micropores with low aspect ratios and the increases in the effect of macro-mesopores (Fig. 6(D)).

In the case condition #2, the local and global parameters were evaluated using the proposed DIA procedure from all magnification data set and Vp measured at an effective pressure of 3.5 MPa (Fig. 7). The main idea is to evaluate the textural parameters under a minimum effective pressure avoiding coupling noise on Vp and Vs measurements. Consequently, we applied the proposed methodology to quantify the microporosity aspect ratio, the minimum error ($e_{\min}$), and the adjustment (R) coefficients as follows:

$$e_{\min} = \frac{|V_{\text{Measured}} - V_{\text{DEM}}|}{V_{\text{Measured}}}, \quad (7)$$

$$R = 1 - e_{\min}. \quad (8)$$

The Vp calculated by applying the DEM model achieved a good agreement with the measured Vp ($R \approx 0.996$) (Fig. 8(A)). The results are considered consistent. Although the Vs has not been used directly in the microporosity aspect ratio prediction methodology, it was calculated

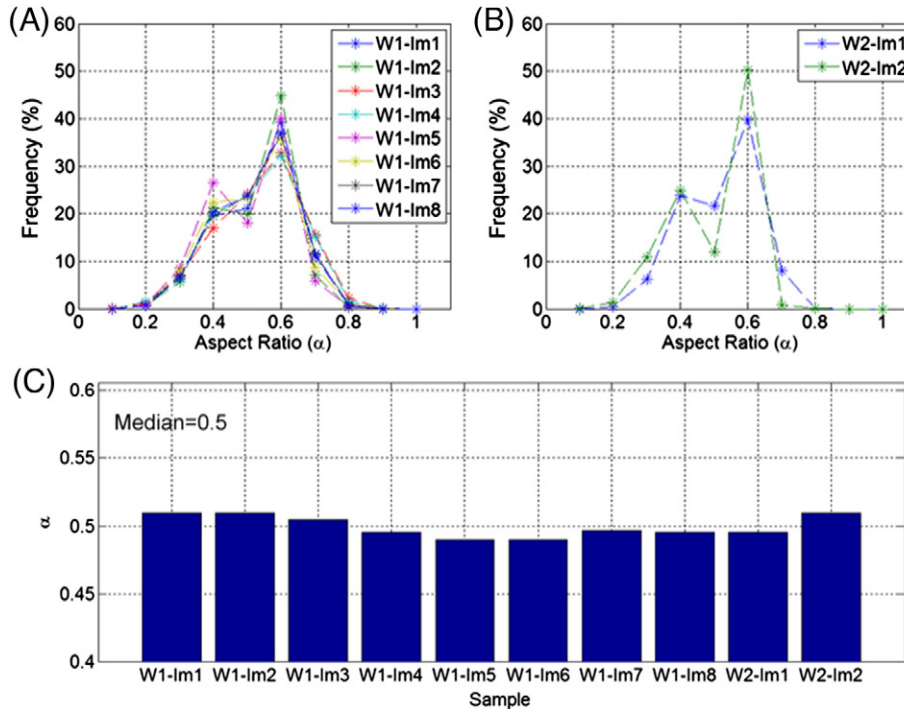


Fig. 5. Results of the Digital Image Analysis (DIA) considering all magnifications available for each core sample (in Appendix A.3). (A) and (B) show the macro-mesopore aspect ratio (α) balance from Wells 1 and 2, respectively, and (C) shows the median aspect ratio of macro-mesopore calculated for each core sample. The representative value for the Albian grainstone core samples is $\alpha = 0.5$.

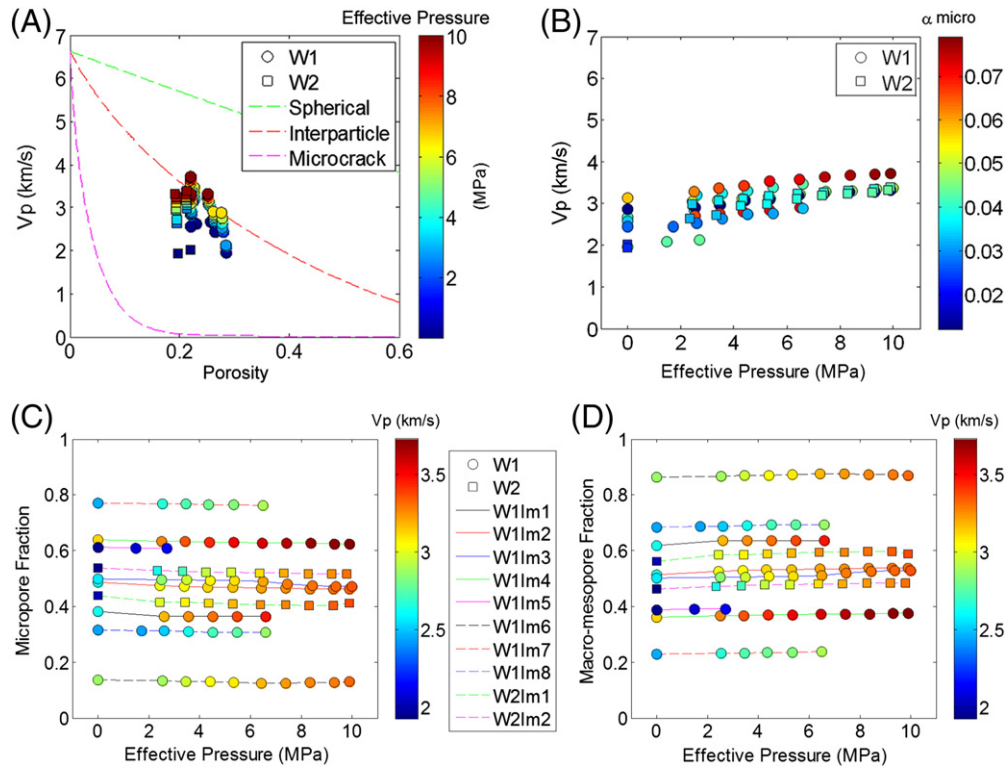


Fig. 6. Textural porosity at all magnification data set and Vp at effective pressure: 0–10 MPa. The results of the analyses reflect the slight porosity volume reduction (~0.4–3.6%) by effective pressure loading. (A) Effective pressure, (B) aspect ratio of the microporosity inclusion — results of the case condition #1, (C) and (D) microporosity and macro-mesopore fractions, respectively, at different Vp results and effective pressures. Spherical, interparticle and microcrack dashed lines are the reference for dry calcite limestone, calculated by the DEM model.

using the bulk and shear moduli calibrated by the pore inclusions, and showed a good agreement with the measurements ($R \approx 0.91$) (Fig. 8(B)). It is a good indication that the methodology works and contributes to the quality control of input parameters and to the evaluation of the methodology efficiency. The methodology can be used to predict Vs in the Albian grainstones according to the obtained results.

The textural impacts were analyzed in case condition #3, using the macro-mesopores from the 25 $\times$ magnification data set and a Vp measured at an effective pressure of 3.5 MPa. The PoA parameter expressed a high sensitivity to the image resolution (Appendix A.3). Thus, we decided to use the thin-section images from the core samples at the lower and representative magnification value of 25 $\times$, assuming image resolution close to the adopted by Weger et al. (2009). In this case, observe that only images of core samples from Well 1 are available (see Appendix A.3) and the PoA ratios were recalculated for the analyses. Fig. 9 shows the P-wave velocity at an effective pressure of 3.5 MPa versus porosity and DOMsize versus PoA. The textural impacts are noted as red arrows according to Weger et al. (2009). The medium aspect ratio and quantity of macro-mesopores ($f_{macro-meso}$) were estimated for analyses of impacts on Vp and textural parameters. Weger et al. (2009) used a large carbonate data set and observed a relationship between the macro-mesopore inclusions that make the rocks stiffer and have higher velocities and the microporosity inclusions that contribute to softer rocks and a lower velocity. Unfortunately, we have a small data set but it is possible to verify that most of data points agree with such behavior, as can be seen in Fig. 9 (C and D).

7. Conclusions

Through a laboratorial analysis of the core samples, the DIA, and the proposed methodology based on the DEM model, the micropore aspect ratios of the inclusions have been predicted.

In this study, the DIA procedure was applied to estimate the local and global parameters, based on the considerations of magnification

and the resulting pixel resolution. The PoA ratio is a good textural parameter, but its variability with image resolution requires caution.

The effective pressure influences the pore system, which mainly affects the microporosity inclusions with respect to the aspect ratio and Vp relationship, and the microporosity is more sensitive at low aspect ratios.

The methodology quantifies the microporosity aspect ratio with the minimum error calibrated from the Vp measured in laboratory. The S-wave velocity (Vs) was not used directly in the proposed methodology to predict the microporosity aspect ratio. However, we observed a high potential and a good agreement for the method of using the microporosity aspect ratio inverted from Vp to predict Vs. It can be helpful to areas where the Vs is unknown.

The analytical theories in the proposed methodology are functions of the macro-mesopore system recognized in the DIA, the amount of microporosity calculated from the difference between the porosities estimated from laboratorial helium-gas and the thin-section petrographic images and the P-wave velocities under dry effective pressure conditions. The results are acceptable, consistent, and suitable for characterizing the Albian grainstone core samples. Other types of heterogeneous rocks could be evaluated using the methodology described in this work. In some cases, the use of other effective elastic media theory instead of the DEM model may produce more realistic results of elastic moduli and velocities.

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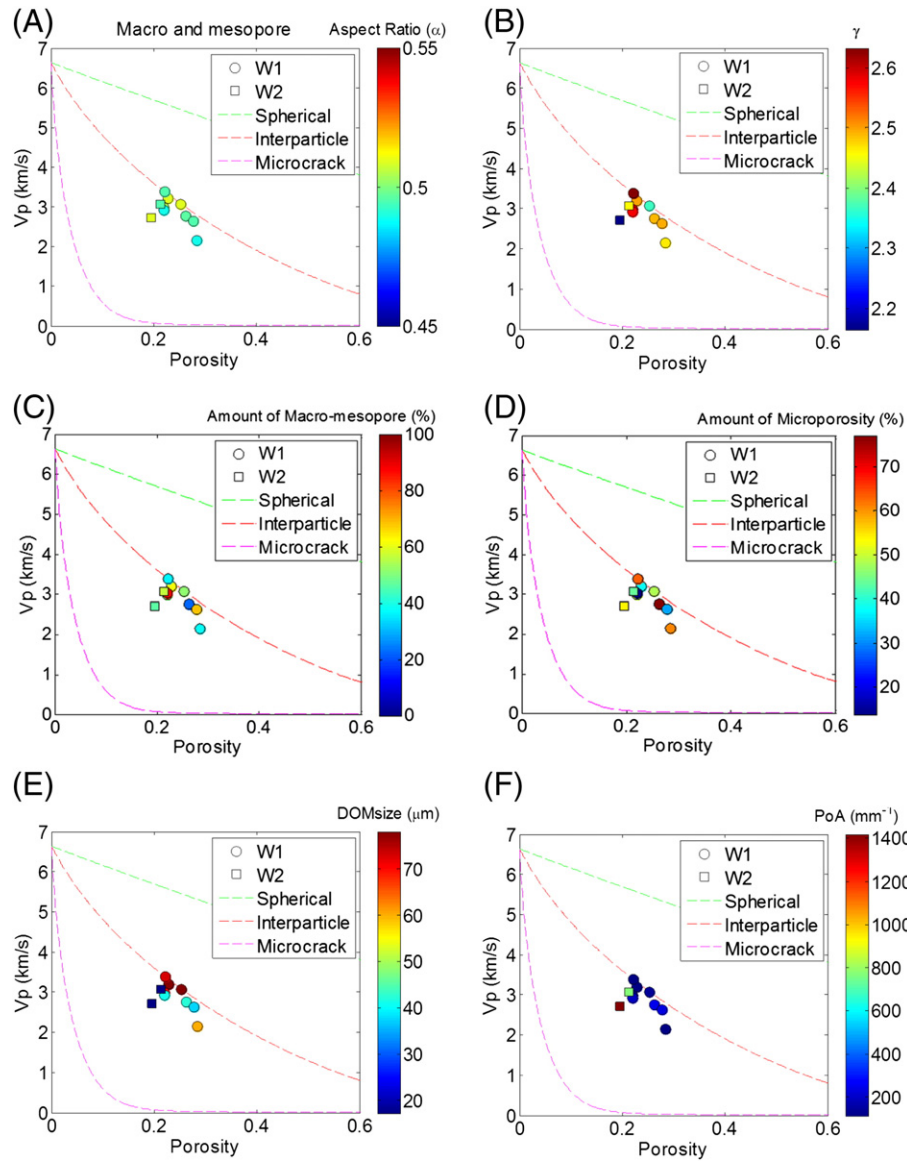


Fig. 7. V_p versus porosity at an effective pressure equal to 3.5 MPa. Each color bar represents a result of the Digital Image Analysis (DIA) parameters from all magnification data sets (case condition #2). Spherical, interparticle and microcrack dashed lines are the reference for dry calcite limestone, calculated by the DEM model.

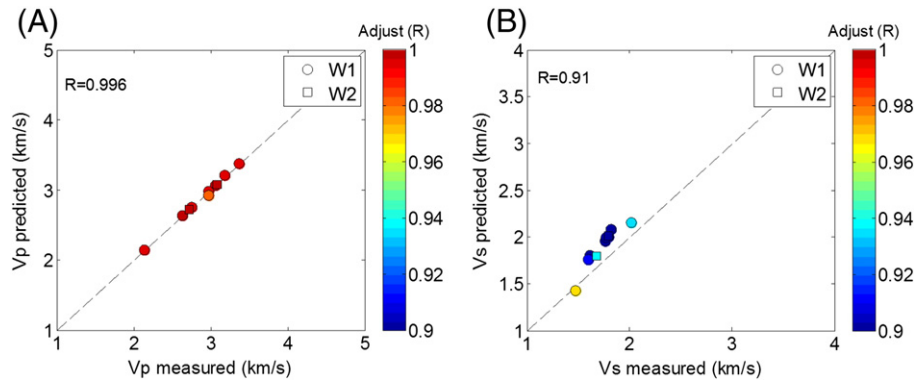


Fig. 8. Results of the proposed methodology to predict the micropore aspect ratios of three pore-space scales in two representative inclusions of the pore system: the macro-mesopores and the microporosity. Results of the case condition #2: (A) and (B) show the adjustment coefficients of the P- and S-wave velocities at effective pressure of 3.5 MPa, respectively, using all magnification data set.

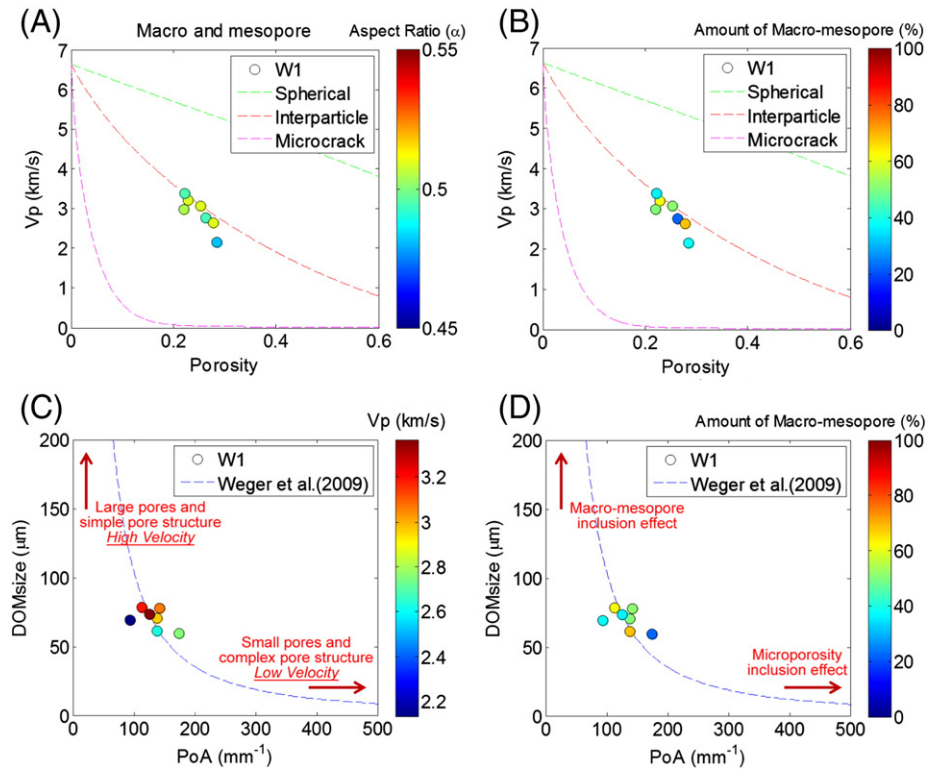


Fig. 9. Results of the case condition #3: effects of the macro-mesopore and microporosity inclusions on the pore structure using Vp at an effective pressure of 3.5 MPa versus porosity, and DOMsize versus PoA for the 25× magnification data set. (A) Macro-mesopore aspect ratio, (B) and (D) proportion of macro-mesopores, (C) Vp. Impacts on velocity are established according to Weger et al. (2009). Spherical, interparticle and microcrack dashed lines are the reference for dry calcite limestone, calculated by the DEM model.

Appendix A. The data set

Table A.1

Petrophysical measurements (part 1 – porosity, permeability, and matrix). The Voigt–Reuss–Hill average method was applied to calculate the effective elastic moduli for a mixture of mineral grains (Mavko et al., 1998).

Sample	Porosity helium-gas (%)	Porosity thin-section (DIA) (%)	Microporosity (%)	Permeability (mD)	Matrix bulk modulus (GPa)	Matrix shear modulus (GPa)	Matrix density (g/cm ³)
W1-lm1	23.03	14.10	8.93	9.0	76.85	32.03	2.710
W1-lm2	25.71	13.00	12.71	221.6	74.65	31.86	2.709
W1-lm3	22.07	11.00	11.07	31.9	76.33	32.09	2.712
W1-lm4	22.28	8.00	14.28	13.5	75.10	31.80	2.709
W1-lm5	28.51	11.00	17.51	126.6	76.47	32.44	2.720
W1-lm6	22.20	19.00	3.20	9.8	76.16	32.16	2.711
W1-lm7	26.42	6.00	20.42	4.4	75.63	31.98	2.710
W1-lm8	27.96	19.00	8.96	8.2	75.90	32.07	2.711
W2-lm1	21.94	12.00	9.94	2.03	76.55	32.08	2.710
W2-lm2	19.72	9.00	10.72	0.88	76.80	32.16	2.711

Table A.2

Petrophysical measurements (part 2 – mineralogical results of XRD and Rietveld method).

Sample	Weight %				
	Calcite	Dolomite	Quartz	Feldspar	Fluorite
W1-01	99.71	0.00	0.29	0.00	0.00
W1-02	95.20	2.13	0.92	1.76	0.00
W1-03	96.98	0.63	1.76	0.63	0.00
W1-04	96.47	1.46	0.52	1.55	0.00
W1-05	95.56	1.19	1.37	0.00	1.88
W1-06	96.98	1.04	1.55	0.43	0.00
W1-07	–	–	–	–	–
W1-08	–	–	–	–	–
W2-01	99.37	0.64	0.00	0.00	0.00
W2-02	98.77	0.38	0.85	0.00	0.00

Table A.3

Data set as a result of the Digital Image Analysis (DIA) procedure.

Sample	Magnification	Aspect ratio		DomSize (μm)	PoA (mm^{-1})	Gamma
		Macro–meso	Micro			
W1-lm1	$\times 25$	0.51	0.04	78.13	112.59	2.50
W1-lm2	$\times 25$	0.51	0.05	77.94	142.85	2.37
W1-lm3	$\times 25$	0.50	0.04	70.67	137.69	2.60
W1-lm4	$\times 25$	0.50	0.07	73.20	125.30	2.63
W1-lm5	$\times 25$	0.48	0.04	68.91	93.07	2.59
	$\times 50$	0.49	0.04	42.50	227.07	2.40
	All	0.49	0.04	60.44	123.78	2.47
W1-lm6	$\times 50$	0.49	0.01	41.25	199.03	2.58
W1-lm7	$\times 25$	0.49	0.07	59.25	173.79	2.54
	$\times 50$	0.50	0.07	35.70	307.51	2.48
	All	0.50	0.07	42.50	251.06	2.49
W1-lm8	$\times 25$	0.51	0.03	61.13	138.33	2.48
	$\times 50$	0.49	0.03	37.65	268.46	2.52
	$\times 100$	0.50	0.03	18.85	488.63	2.47
	All	0.50	0.03	37.37	240.97	2.49
W2-lm1	$\times 100$	0.49	0.04	17.83	763.81	2.51
	$\times 200$	0.51	0.04	9.50	966.36	2.35
	All	0.50	0.04	17.25	773.98	2.46
W2-lm2	$\times 100$	0.51	0.03	17.39	1418.31	2.16

Table A.4

Data set as a result of the Digital Image Analysis (DIA) procedure for all magnifications and P- and S-wave at an effective pressure of 3.5 MPa.

Sample	Vp (km/s)	Vs (km/s)	Aspect ratio		Adjust coefficient R (fraction)	DomSize (μm)	PoA (mm^{-1})	Gamma
			Macro–meso	Micro				
W1-lm1	3.186	1.824	0.51	0.04	0.994	78.13	112.59	2.50
W1-lm2	3.061	1.796	0.51	0.05	0.998	77.94	142.85	2.37
W1-lm3	2.969	1.766	0.50	0.04	0.995	70.67	137.69	2.60
W1-lm4	3.367	2.016	0.50	0.07	0.996	73.20	125.30	2.63
W1-lm5	2.135	1.475	0.49	0.04	0.996	60.44	123.78	2.47
W1-lm6	2.973	1.771	0.49	0.01	0.983	41.25	199.03	2.58
W1-lm7	2.751	1.616	0.50	0.07	0.997	42.50	251.06	2.49
W1-lm8	2.635	1.597	0.50	0.03	1.000	37.37	240.97	2.49
W2-lm1	3.077	1.798	0.50	0.04	0.999	17.25	773.98	2.46
W2-lm2	2.723	1.680	0.51	0.03	1.000	17.39	1418.31	2.16

Appendix B. Geometric coefficients $P^{(*2)}$ and $Q^{(*2)}$ for ellipsoidal inclusions of arbitrary aspect ratios

For dry ellipsoidal pore inclusions, the geometric coefficients are given by the following (Berryman, 1980; Mavko et al., 1998):

$$P = \frac{1}{3} T_{ijj}, \quad (\text{B.1})$$

$$Q = \frac{1}{5} \left(T_{ijj} - \frac{1}{3} T_{iii} \right), \quad (\text{B.2})$$

where the tensor T_{ijkl} relates the uniform strain field to the strain field within the ellipsoidal inclusion (Wu, 1966). Berryman (1980) gave the scalar formulation required for calculating P and Q as follows:

$$T_{ijj} = \frac{3F_1}{F_2}, \quad (\text{B.3})$$

$$T_{ijj} = \frac{F_1}{F_2} + \frac{2}{F_3} + \frac{1}{F_4} + \frac{F_4 F_5 + F_6 F_7 - F_8 F_9}{F_2 F_4}, \quad (\text{B.4})$$

where

$$F_1 = 1 + A \left[\frac{3}{2} (f + \theta) - R \left(\frac{3}{2} g + \frac{5}{2} \theta - \frac{4}{3} \right) \right], \quad (\text{B.5})$$

$$F_2 = 1 + A \left[1 + \frac{3}{2} (f + \theta) - \frac{R}{2} (3f + 5\theta) \right] + B(3 - 4R), \quad (\text{B.6})$$

$$+ \frac{A}{2} (A + 3B)(3 - 4R) \left[f + \theta - R(f - \theta + 2\theta^2) \right]$$

$$F_3 = 1 + A \left[1 - \left(f + \frac{3}{2} \theta \right) + R(f + \theta) \right], \quad (\text{B.7})$$

$$F_4 = 1 + \frac{A}{4} [3\theta + f - R(f - \theta)], \quad (\text{B.8})$$

$$F_5 = A \left[R \left(f + \theta - \frac{4}{3} \right) - f \right] + B\theta(3 - 4R), \quad (\text{B.9})$$

$$F_6 = 1 + A[1 + f - R(f + \theta)] + B(1 - \theta)(3 - 4R), \quad (\text{B.10})$$

$$F_7 = 2 + \frac{A}{4}[9\theta + 3f - R(5\theta + 3f)] + B\theta(3 - 4R), \quad (\text{B.11})$$

$$F_8 = A\left[1 - 2R + \frac{f}{2}(R - 1) + \frac{\theta}{2}(5R - 3)\right] + B(1 - \theta)(3 - 4R), \quad (\text{B.12})$$

$$F_9 = A[f(R - 1) - R\theta] + B\theta(3 - 4R), \quad (\text{B.13})$$

$$A = \frac{\mu_i}{\mu_m} - 1, \quad (\text{B.14})$$

$$B = \frac{1}{3} \left(\frac{K_i}{K_m} - \frac{\mu_i}{\mu_m} \right), \quad (\text{B.15})$$

$$R = \frac{3\mu_m}{3K_m + 4\mu_m}, \quad (\text{B.16})$$

$$\theta = \begin{cases} \frac{\alpha}{(1 - \alpha^2)^{3/2}} \left[\cos^{-1} \alpha - \alpha(1 - \alpha^2)^{1/2} \right], & (\alpha > 1) \\ \frac{\alpha}{(\alpha^2 - 1)^{3/2}} \left[\alpha(\alpha^2 - 1)^{1/2} - \cosh^{-1} \alpha \right], & (\alpha < 1) \end{cases}, \quad (\text{B.17})$$

$$f = \frac{\alpha^2}{1 - \alpha^2} (3\theta - 2), \quad (\text{B.18})$$

where α is the aspect ratio of the inclusion; K_m and μ_m are the bulk and shear moduli of the rock matrix, respectively; and K_i and μ_i are the bulk and shear moduli of the inclusion, respectively.

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